UniForge: Unified Engineering Workbench

Candidate ID: RA2311042010060 · **Name:** Aakash G S · **Repository:** github.com/ftaakash/UniForge (tag `v1.0.0`) · **Package:** `RA2311042010060_Workbench.zip` · 2 November 2026

> One Windows application with six views (Overview, Geometry, Thermal, Performance, Hardware, Backprop) covers tasks A–F. It starts with one double-click on `Launch.exe`, needs no installer or runtime, and is ready in about 0.13 s. Every number in this report is generated by the shipped tool `app\uniforge_cli.exe` and committed under `results/`; 289 automated tests pass on Windows (MSVC) and the portable core also builds and passes under ASan/UBSan and TSan on Linux (GCC 13) in CI.

1. Design choices (Task A)

- **C++17 native application.** Dear ImGui (docking) + ImPlot on Win32 and Direct3D 11, with a WARP software fallback for machines without a GPU driver. The MSVC runtime is linked statically and shaders are precompiled, so the executables import only DLLs that ship with Windows. `tools/package.ps1` checks this with `dumpbin /dependents`.
- **Three layers.** `core/` is portable standard C++ with no UI or OS headers (thermal, io, perf, hw, ml); `services/` runs work off the UI thread; views only draw. The same core powers the GUI and the CLI, so the GUI shows exactly what the evidence files contain.
- **Background jobs.** One job thread with progress, cooperative cancellation and an atomic `shared_ptr` publish: the UI never blocks, and a cancelled or failed job keeps the previous result.
- **Errors are values.** Every fallible call returns `Result<T>` with an `ErrorCode`, a message and a hint (for files: the line number).
- **Robust delivery.** Paths with spaces and non-ASCII characters work; a read-only install folder moves logs to `%LOCALAPPDATA%`; a crash handler writes `logs/crash_<pid>.txt`. All three were tested on the package (`results/edge_cases.md`).
- **Evidence by construction.** Results are written atomically with `%.17g` precision and a `.meta.json` sidecar (version, commit, inputs); golden-file tests prove the CLI regenerates them byte for byte.

2. CAD / ANSYS data (Task B)

A streamed line reader (64 KB buffer; LF/CRLF; UTF-8 BOM) feeds an ASCII STL parser and a nodal-temperature CSV parser. Both validate every value (non-finite, overflow, vertex count, duplicate node ids, temperatures below absolute zero) and fail with a typed error naming the line. 37 malformed fixtures are covered by negative tests. Limits are enforced before memory is committed and were measured on the target machine:

File	Elements	Size (MB)	Parse (s)	MB/s	Peak memory (MB)
ASCII STL	1 000 000	248.0	0.998	248	145
ASCII STL (limit)	2 000 000	496.0	2.008	247	324
CSV nodes (limit)	5 000 000	274.9	1.340	205	386

The Geometry view renders the mesh (flat shading, optional wireframe, MSAA 4×) and the nodes as an inferno-coloured point cloud with a °C legend; clicking picks the nearest node within 6 px. The sample rod STL is generated geometry; the temperature CSV is **synthetic** and labelled as such in the file name, the UI and `data/PROVENANCE.md`.

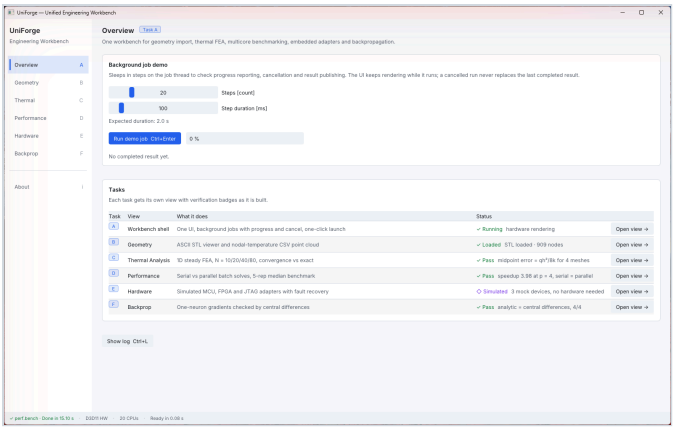


Figure 1. Overview: one row per task with live status.

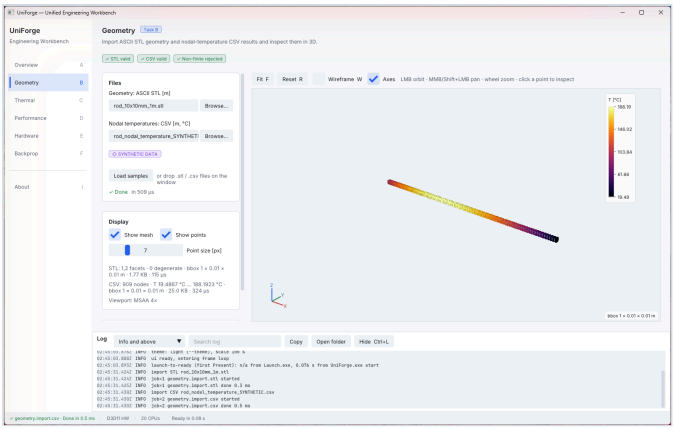


Figure 2. Geometry: sample rod and synthetic nodal temperatures.

3. Thermal FEA (Task C)

Steady 1D conduction, `−kA·T″ = qA` on 0 ≤ x ≤ L with T(0) = 100 °C, T(L) = 20 °C, L = 1 m, A = 1e-4 m², k = 10 W/(m·K), q = 1e4 W/m³. Two-node linear elements give k/h·[1 −1; −1 1] and qh/2·[1 1]; the global matrix is tridiagonal and symmetric. Dirichlet values are applied by **symmetric elimination** (known temperatures moved to the right-hand side), so the reduced matrix stays SPD and irreducibly diagonally dominant and the **Thomas algorithm** solves it in O(N) without pivoting. The exact solution T(x) = T0 + (TL − T0)x/L + q x(L − x)/(2k) peaks at 188.2 °C at x = 0.42 m.

N	h (m)	max midpoint error (°C)	$qh^2/8k$ (°C)	ratio	order	max nodal error (°C)	energy residual (W)
10	0.1	1.250000	1.25	–	–	2.8e-14	2.2e-16
20	0.05	0.312500	0.3125	4.0000	2.0000	2.0e-13	1.1e-15
40	0.025	0.078125	0.078125	4.0000	2.0000	8.8e-13	5.1e-15
80	0.0125	0.019531	0.01953125	4.0000	2.0000	1.2e-12	8.7e-15

Why these numbers. For a quadratic exact solution, linear elements are nodally exact in 1D, so nodal errors are round-off only. Between nodes the FE solution is the linear interpolant, whose error at each midpoint is exactly $q\,h^2/(8k)$: halving h divides it by 4 (order 2). The energy balance (heat generated = heat leaving both ends, from the unconstrained boundary rows) closes to 1e-14 W. These are checked by 61 numerical tests, including a dense Gaussian-elimination oracle. A **deliberate fault injection** (dropping the T0 contribution in the boundary elimination) still passed the boundary-value checks but failed 18 tests and showed an energy residual equal to the missing term, which is why the energy balance is a release check (`docs/thermal_fea.md` §5, debugging log).

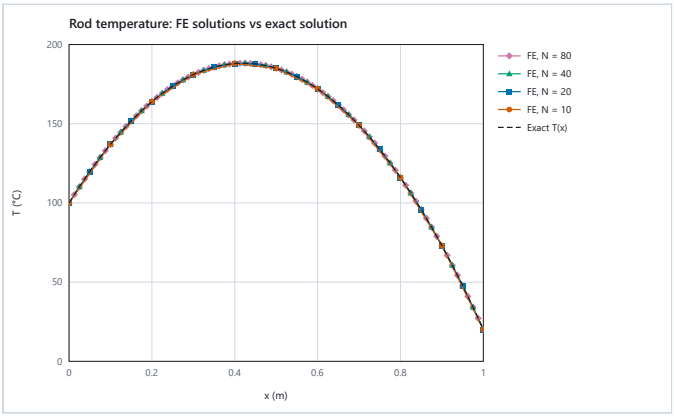


Figure 3. FE solutions for N = 10–80 and the exact T(x).

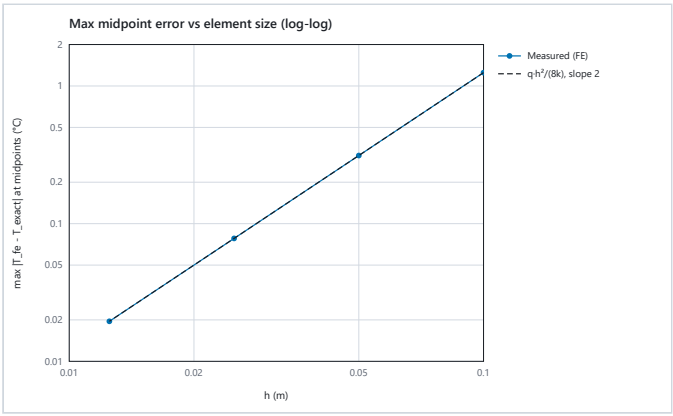


Figure 4. Max midpoint error vs h on a log–log scale: slope 2.

4. Multicore computing (Task D)

A batch of independent rod problems (k, q drawn from a portable seeded RNG so the batch is identical on MSVC and GCC) is solved serially and with a bounded pool of p `std::thread`s. Workers take chunks of max(1, cases/(8p)) from an atomic index and write only their own result slots, so there is no locking and results stay in case order. Protocol: 1 warm-up + 5 timed repetitions per p, median reported with min–max; serial and parallel results are compared on every node of every repetition.

p	median (s)	min–max (s)	speedup	efficiency
1	1.0732	1.0700–1.0742	1.00	100 %
2	0.5353	0.5337–0.5386	2.00	100 %
4	0.2686	0.2661–0.2746	4.00	100 %

10 000 cases × 8 000 elements, seed 20261101; Intel i5-14600K (6 P + 8 E cores, 20 logical CPUs), Windows 11, MSVC 19.44 /O2. max |ΔT| serial vs parallel = 0 °C (bitwise equal).

Why near-ideal. Each case is O(N) work with no communication; the only shared state is the chunk counter (32 chunks at p = 4). The per-worker working set (~0.45 MB) fits in a P-core's 2 MB L2, and the only DRAM stream is the 640 MB result write (~2.4 GB/s at p = 4), so the run is compute-bound. **Limits:** single-core turbo makes T1 relatively fast (understating efficiency); under load a worker may land on an E-core (an earlier loaded run gave S(4) = 3.88); beyond 6 threads hyper-threading and E-cores would lower efficiency for reasons unrelated to the code. Thread start/join costs < 0.1 % here, which is why the small preset is not used for the report.

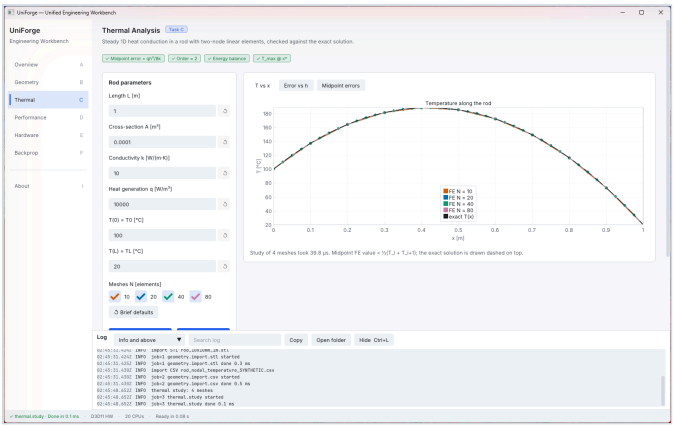


Figure 5. Thermal view: inputs with ranges, four verification badges, plot tabs.

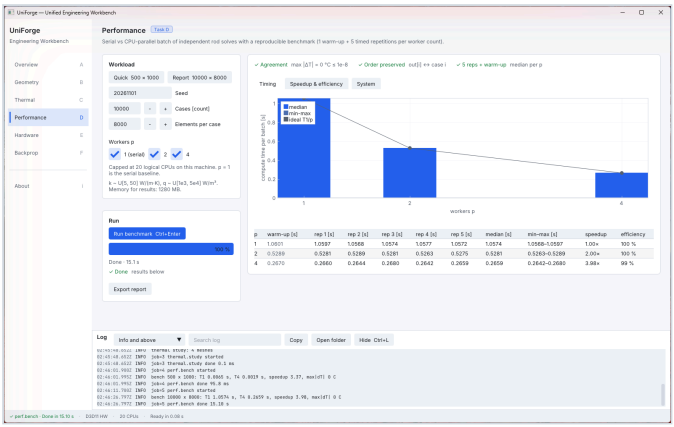
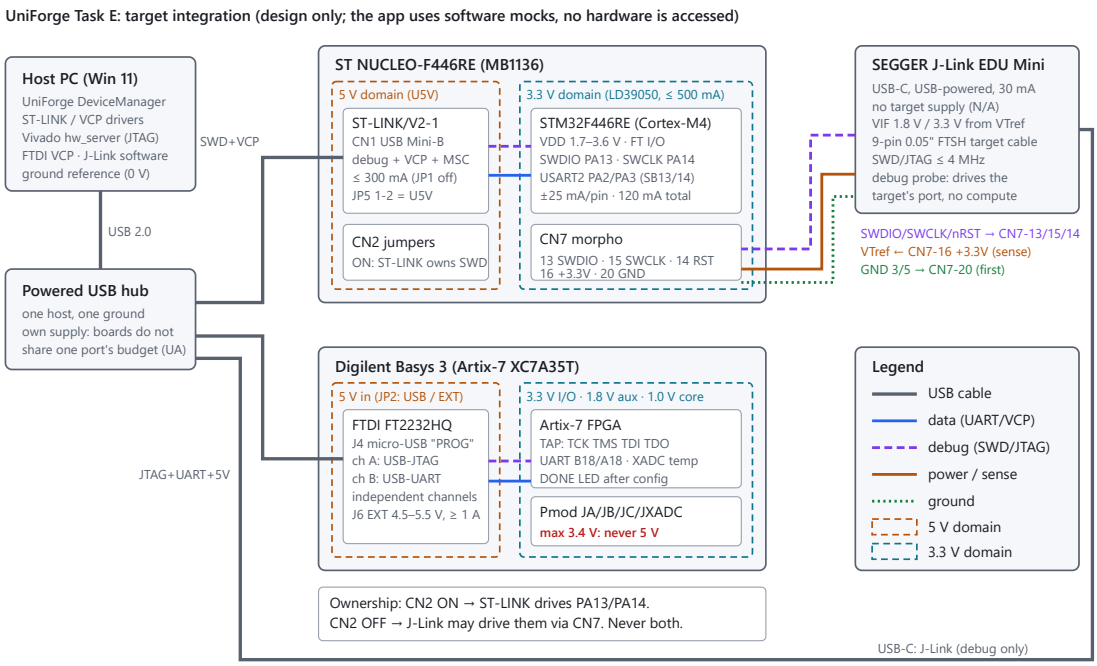


Figure 6. Performance view after a live report-preset run.

5. Embedded integration (Task E)

Design for a host PC talking to an **ST NUCLEO-F446RE** (ST-LINK/V2-1: SWD + virtual COM port), a **Digilent Basys 3** (FT2232HQ: USB-UART + USB-JTAG) and a **SEGGER J-Link EDU Mini** (SWD to the Nucleo's STM32F446). `docs/embedded_design.md` gives the power, voltage-domain, I/O-level, connector and debug-signal table with a citation for every value (user manuals and datasheets S1–S6; unverified assumptions marked UA), a block diagram, and the safety rules: common ground first, never drive a Basys 3 Pmod above 3.4 V (no 5 V Nucleo signal to a Pmod without a level shifter), the J-Link supplies no target power and follows VTref.

The C++ side defines one `IDeviceAdapter` interface with a state machine (Disconnected → Connecting → Connected → Faulted → Recovering) and a retry policy (250 ms timeout, 3 retries after 100/200/400 ms). The three adapters run on a mock transport with injectable faults (timeout once/always, disconnect); **no hardware is accessed** and every card is labelled SIMULATED. Telemetry is polled at 10 Hz on a device thread; recovery is logged ("recovered after 1 retry").



Sources: ST UM1724 Rev 17, ST DS10693 Rev 11, Digilent Basys 3 Reference Manual, SEGGER J-Link EDU Mini spec + KB. UA = unverified assumption. See embedded_design.md.

Figure 7. Block diagram: host PC, NUCLEO-F446RE, Basys 3 and J-Link EDU Mini, with the voltage domains and the links used (full table and citations in `docs/embedded_design.md`).

6. Backpropagation (Task F)

Network: $z = wx + b$, $h = \tanh z$, $\hat{y} = vh + c$, $J = \tfrac{1}{2}(\hat{y} - y)^2$ with $x = 1$, $y = 2$, $(w, b, v, c) = (0.5, 0, 1, 0)$, learning rate 0.1. With $e = \hat{y} - y$: $\partial J/\partial v = e \cdot h$, $\partial J/\partial c = e$, $\partial J/\partial w = e \cdot v \cdot (1 - h^2) \cdot x$, $\partial J/\partial b = e \cdot v \cdot (1 - h^2)$. The full derivation is in my **handwritten appendix** (`docs/appendix_handwritten/`).

param	analytic	central difference (ε = 1e-6)	|diff|	≤ 1e-5
w	−1.2094645	−1.2094645	1.1e-10	pass
b	−1.2094645	−1.2094645	1.1e-10	pass
v	−0.7106820	−0.7106820	5.3e-11	pass
c	−1.5378828	−1.5378828	1.1e-10	pass

Training with simultaneous updates drives J from 1.1825 to 1.7e-20 in 100 steps; on a log scale the decay is linear (geometric convergence), since one sample with four parameters can be fitted exactly.

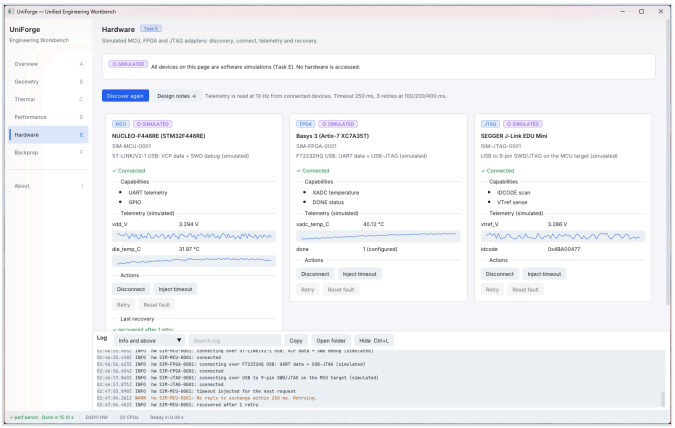


Figure 8. Hardware: three simulated devices; the MCU recovered from an injected timeout.

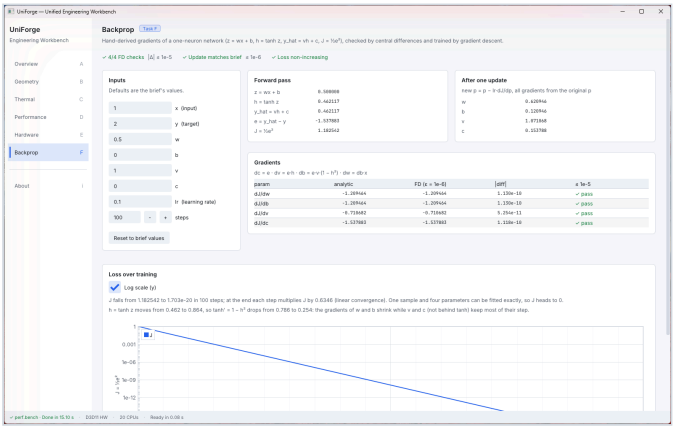


Figure 9. Backprop: forward pass, update, gradient check, loss curve.

7. Testing and quality

Suite	Tests	Covers
unit	145	writers, RNG, element matrices, parsers, camera/picking, formatting, adapters, job runner, worker pool
numerical	61	FEA vs exact and dense oracle, convergence order, energy balance; serial = parallel; backprop values and FD check
negative	69	37 malformed STL/CSV fixtures, invalid FEA and backprop inputs, device faults and recovery
integration	14	CLI regenerates committed evidence byte for byte; import → render-model pipeline; study job and exports

CI (GitHub Actions) builds with `/W4 /WX` on windows-2022 and runs the core under ASan/UBSan and TSan on ubuntu-24.04. The package script runs the tests, the dependency check and a hygiene scan (no absolute paths, e-mail addresses or secrets in 485 files). Manual testing of the package found three UI defects that the tests missed (an ImGui ID clash, a clipped progress bar, and action buttons disabled by the telemetry tick); all were fixed before release.

8. Limitations

- Hardware is simulated; the pin/voltage design is cited but not tested on real boards.
- The temperature CSV is synthetic; there is no real ANSYS export in the repository. Binary STL is detected and rejected with a hint.
- The FEA is 1D, steady and linear (constant k, uniform q), as the brief specifies.
- Not checked: 200 % DPI, full keyboard-only use, high contrast; no screen-reader support (Dear ImGui exposes no UI Automation tree).
- Executables are unsigned, so SmartScreen may warn on first launch.
- Benchmarks are from one machine (i5-14600K); p = 8/16 were not measured.

9. AI use, ownership and effort

AI assistance (Claude Code, Claude Opus 5.5) was used for analysis, code, tests and documentation, under my direction: I chose the design from three alternatives, set and re-planned the phases, reviewed every phase before it reached `main`, and tested the package myself. The handwritten derivation is my own work. Details, including incorrect AI output and how it was caught, are in `docs/ai_usage_log.md`; original vs reused work is in `docs/ownership.md`; hours per phase are in `docs/effort_log.md`.